Purpose

Step-by-step procedure on creating Certificate Templates that will be compatible with our non-Windows devices of varying capabilities.

Scope

In-Scope

- Creation of Certificate Template with 2048-bit key and SHA 256 for Non-Windows devices
- Creation of Certificate Template with 2048-bit key and SHA 1 for Non-Windows devices
- Creation of Certificate Template with 1024-bit key and SHA 1 for Non-Windows devices

Out-of-Scope

- Manual install of certificates on non-domain joined or non-Windows devices

Requirements

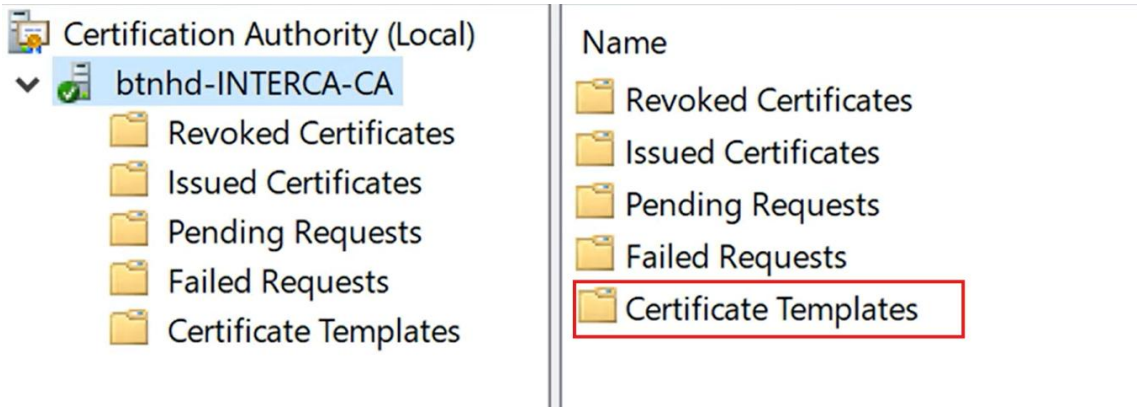
- Permissions
    - Domain Administrator on Intermediate CA
- Remote Access to Intermediate CA server

Procedure

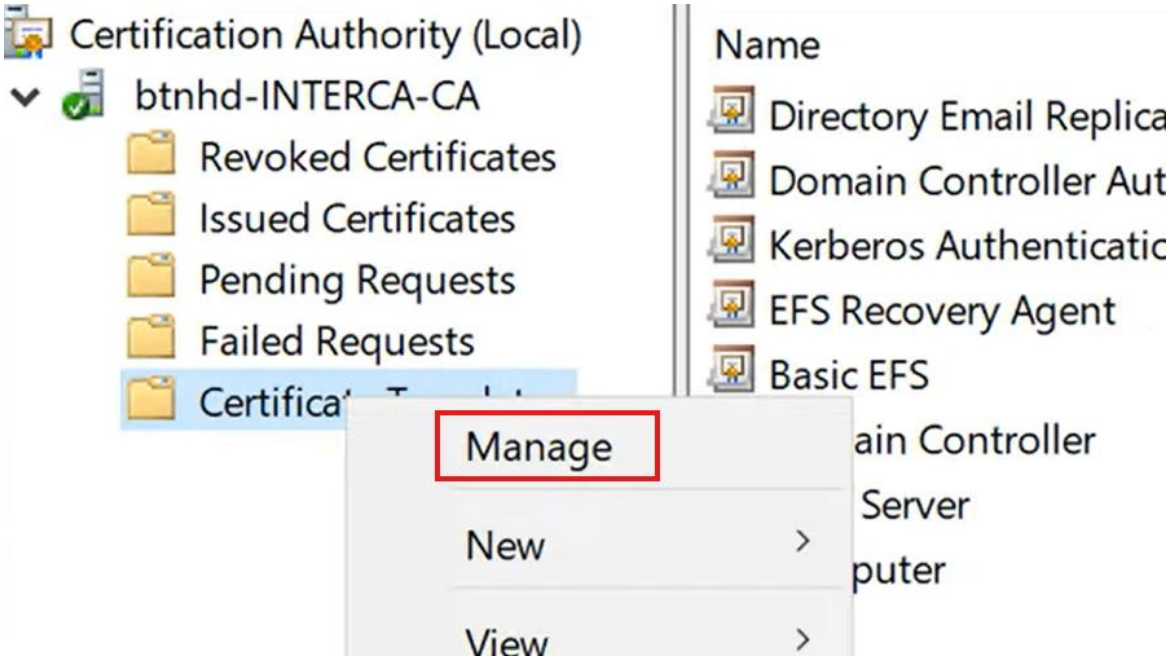
[High-Security] Creation of SHA256-2048bit Non-Windows Certificate Template

In the Intermediate CA server

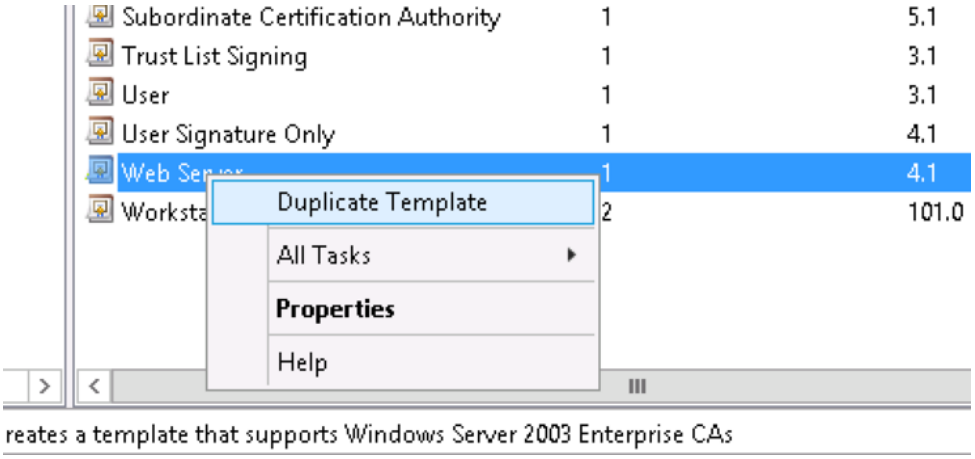
1. Open **Certification Authority (certsrv.msc)**
2. Under our server, go to **Certificate Templates**



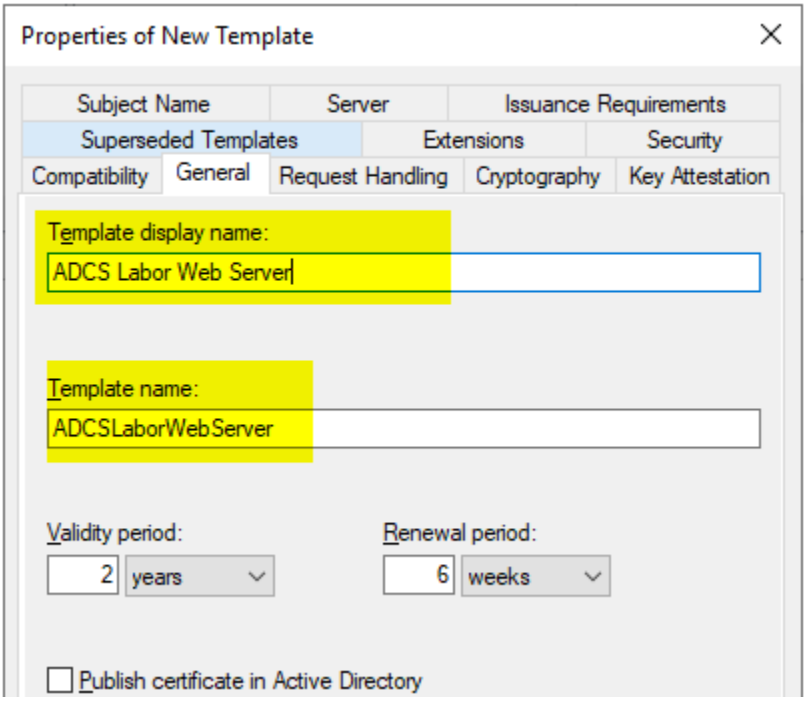
3. Right click **Certificate Templates** and click **Manage**



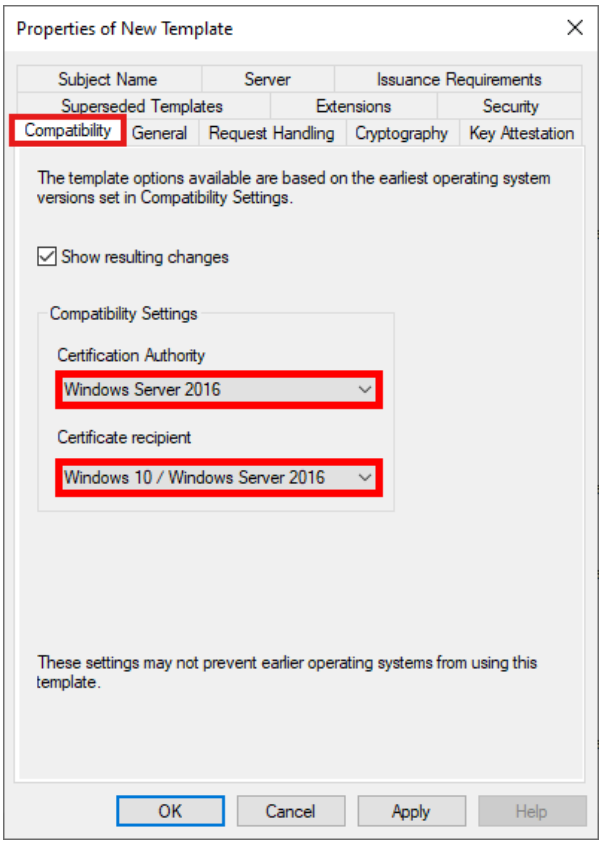
4. Right-click **Web Server** and click **Duplicate Template**



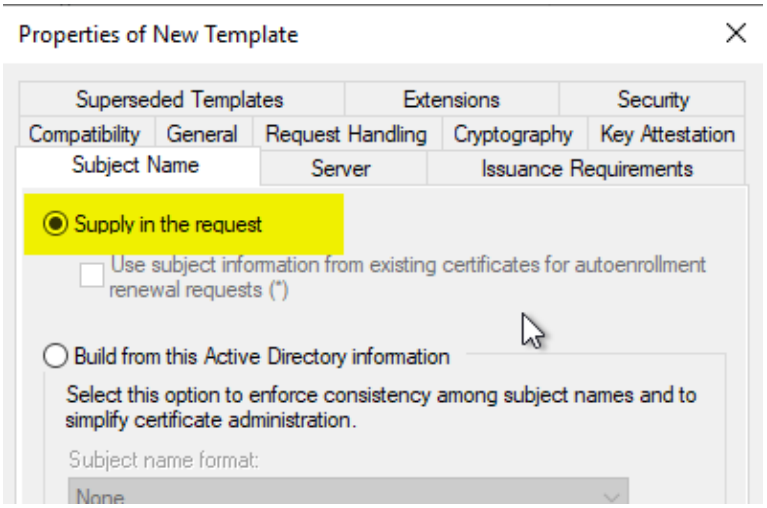
5. In the **General** tab,
   a. Name the template as necessary. I suggest **SHA256-2048 NonWindows Authentication**
   b. Set the **Validity Period** as necessary (preferably **2-3 years**) Some web browsers might reject if validity is longer than 398 days, if so change the Validity Period to 1 year



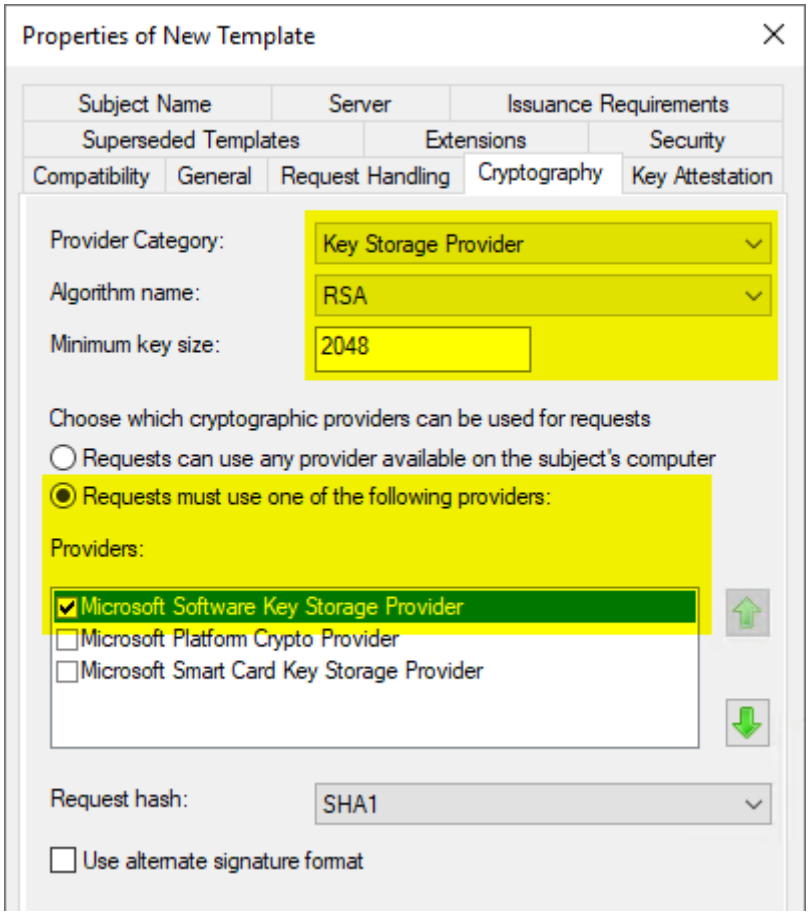
6. In the **Compatibility** tab,
    a. Set **Certification Authority** as **Windows Server 2008 R2**
    b. Set **Certification Recipient** as **Windows 7 / Server 2008 R2**



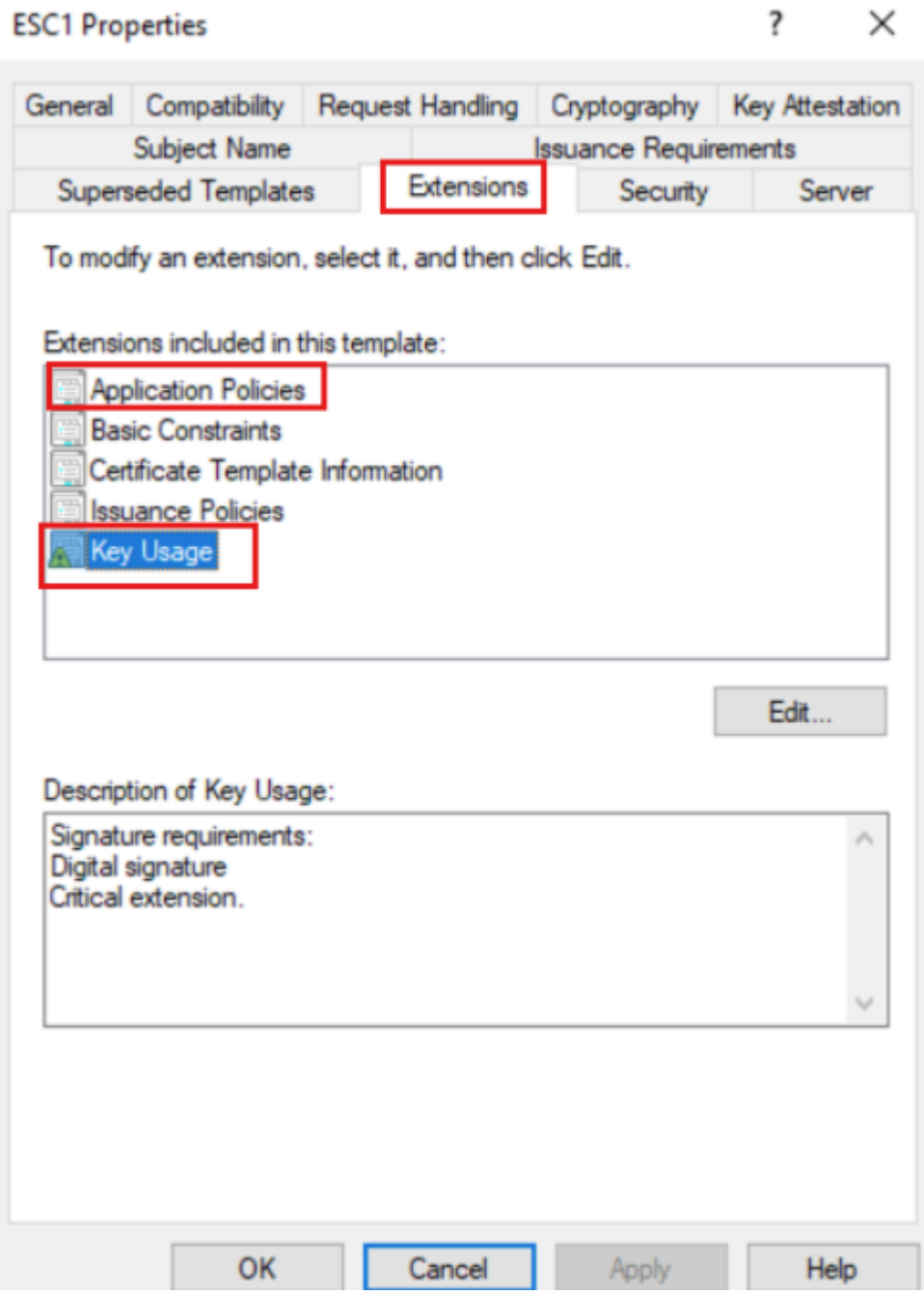
7. In the **Subject Name** tab,
    a. Select **Supply in the request**



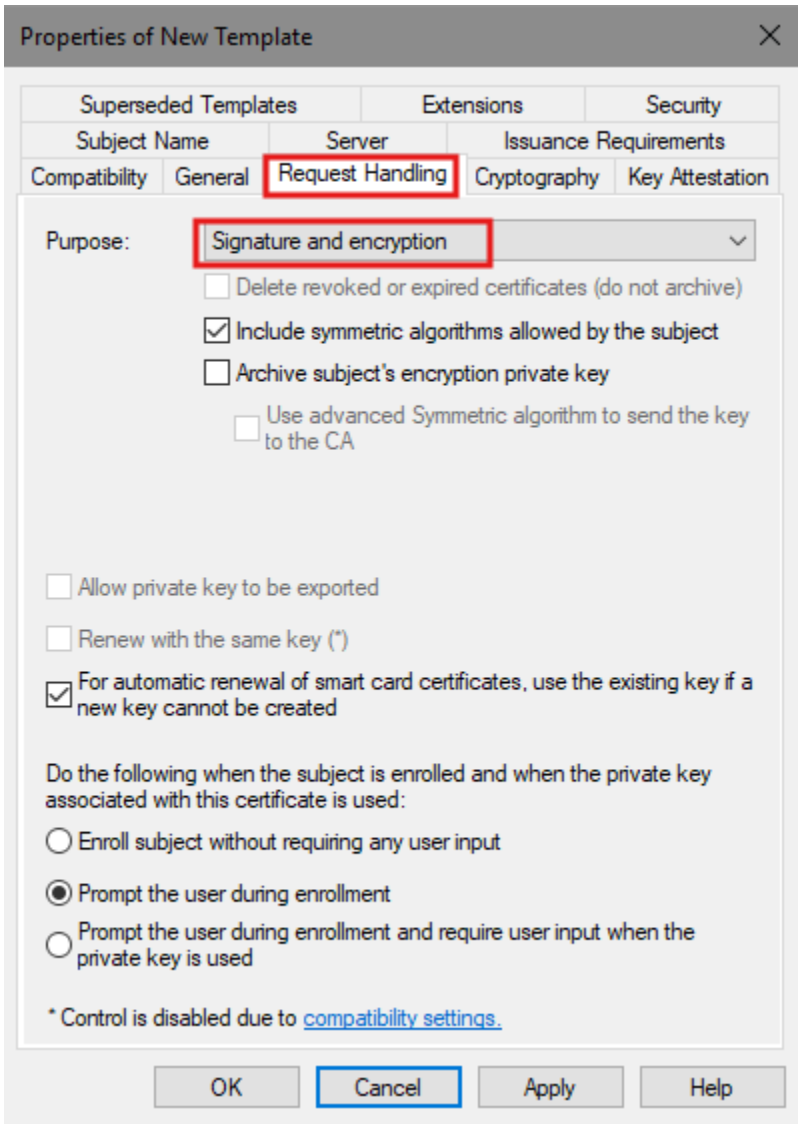
8. In the **Cryptography** tab,
    a. Set **Provider Category** as **Legacy Cryptographic Service Provider**
    b. Set **Algorithm Name** as **RSA**
    c. Set **Minimum Key Size** as **2048**
    d. Check **Requests must use one of the following providers:**
    e. For **Providers**, check **Microsoft RSA Schannel Cryptographic Provider**
    f. For **Request hash**, set to **SHA256**



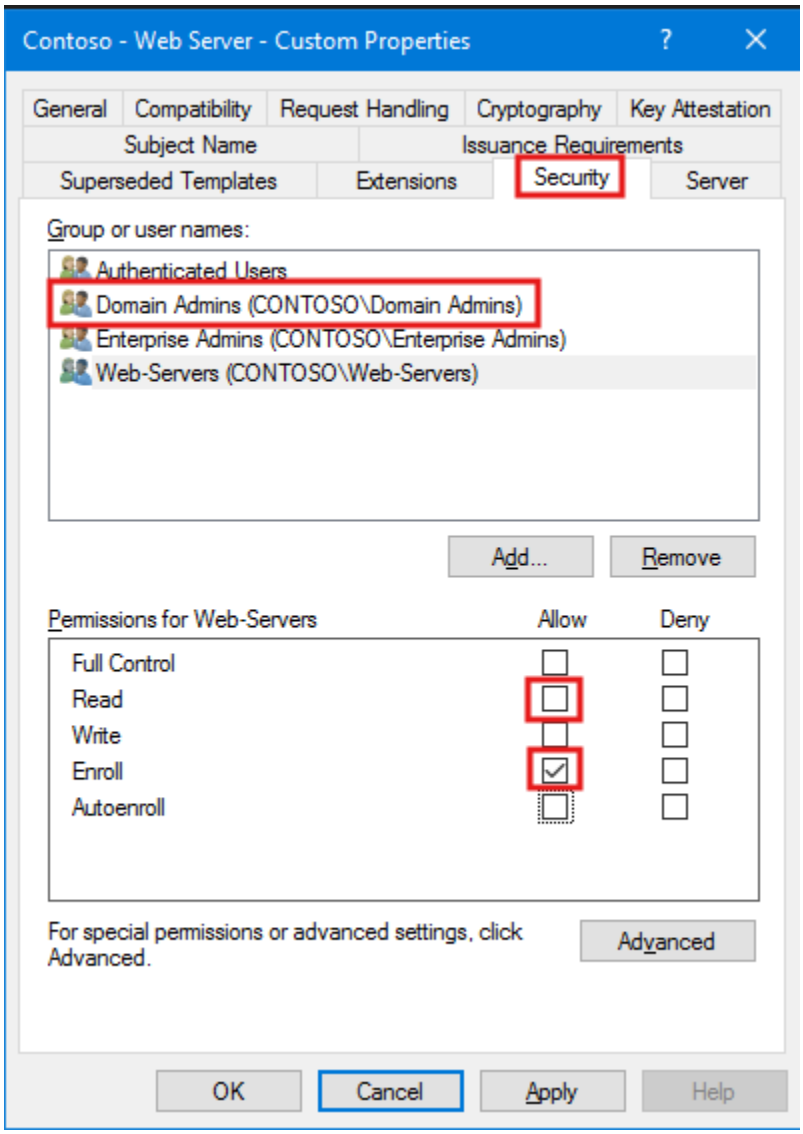
9. In the **Extension** tab,
    a. For **Application Policies**, ensure **Server Authentication** is listed
    b. For **Key Usage**, ensure **Digital Signature** and **Key Encipherment** are listed



10. In the **Request Handling** tab,
    a. Set **Purpose** to **Signature and encryption**
    b. Uncheck **Allow private key to be exported** (for security)



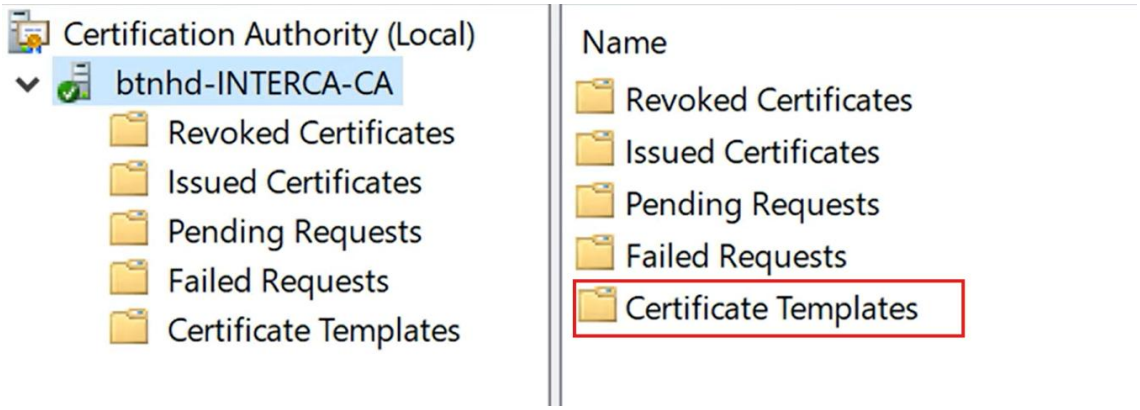
11. In the **Security** tab,
    a. For any group, such as Domain Admins, that will be allowed to enroll/issue certs on InterCA, ensure the permissions, **Read** and **Enroll** are checked to **Allow**



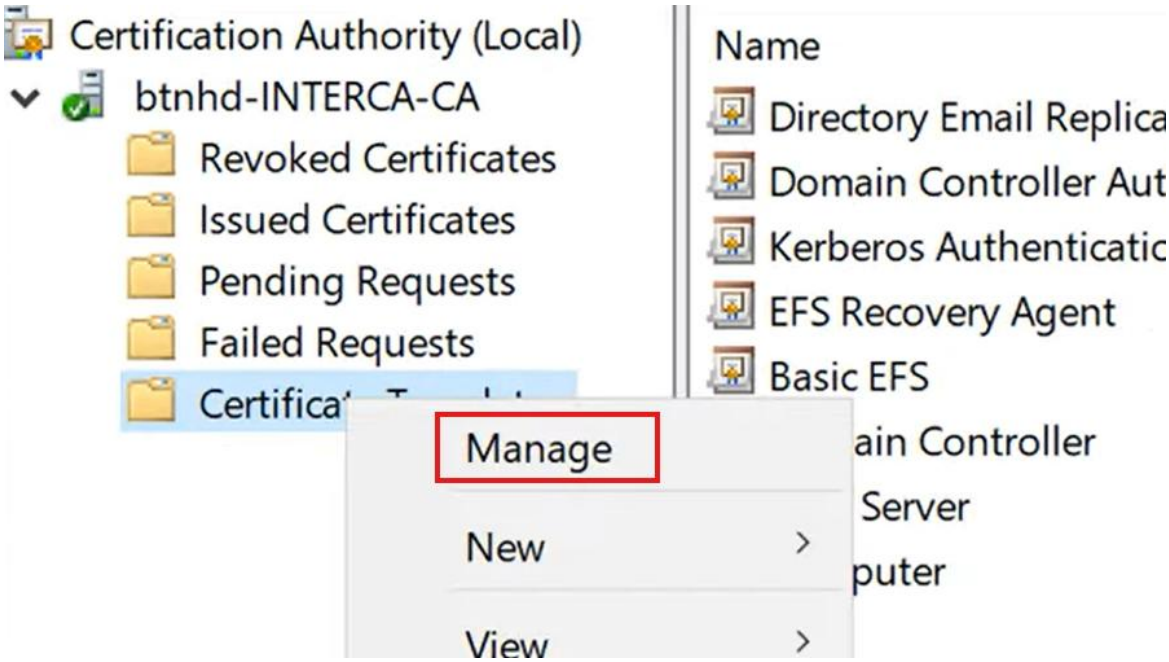
[Med-Security] Creation of SHA1-2048bit Non-Windows Certificate Template

In the Intermediate CA server

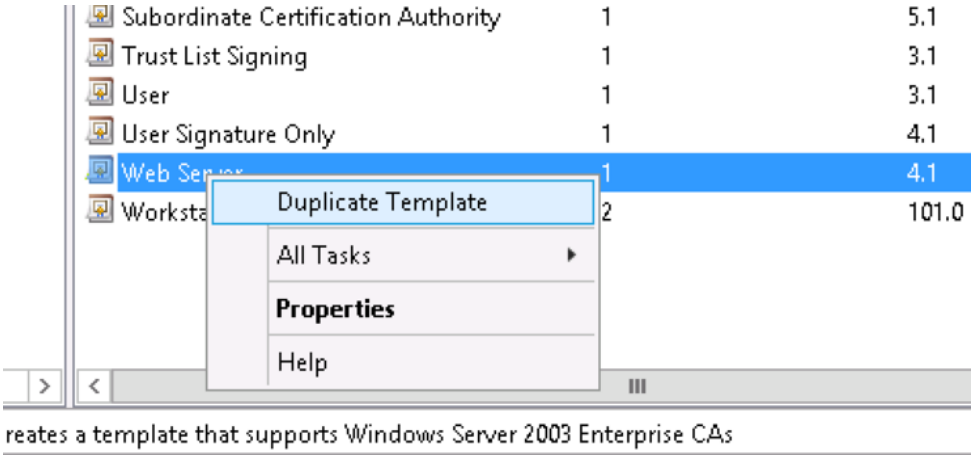
1. Open **Certification Authority (certsrv.msc)**
2. Under our server, go to **Certificate Templates**



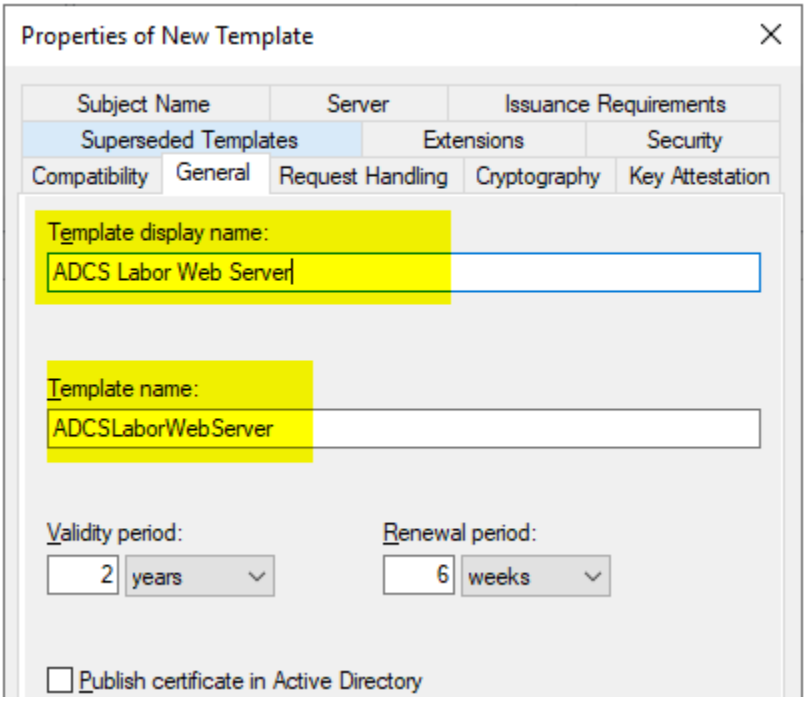
3. Right click **Certificate Templates** and click **Manage**



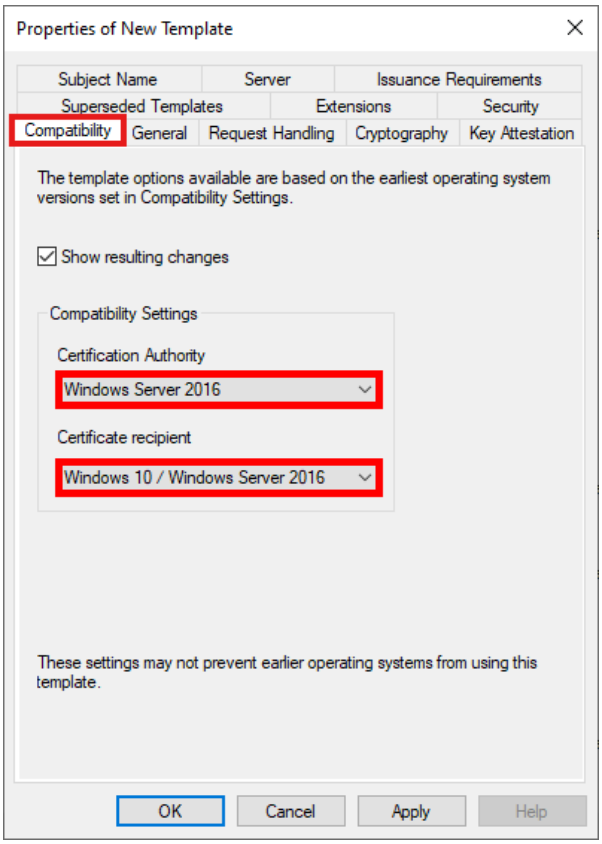
4. Right-click **Web Server** and click **Duplicate Template**



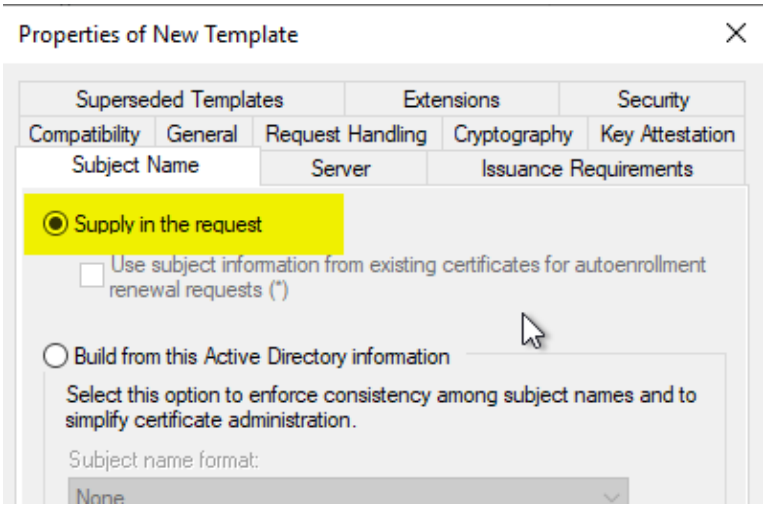
5. In the **General** tab,
   a. Name the template as necessary. I suggest **SHA1-2048 NonWindows Authentication**
   b. Set the **Validity Period** as necessary (preferably **2-3 years**) Some web browsers might reject if validity is longer than 398 days, if so change the Validity Period to 1 year



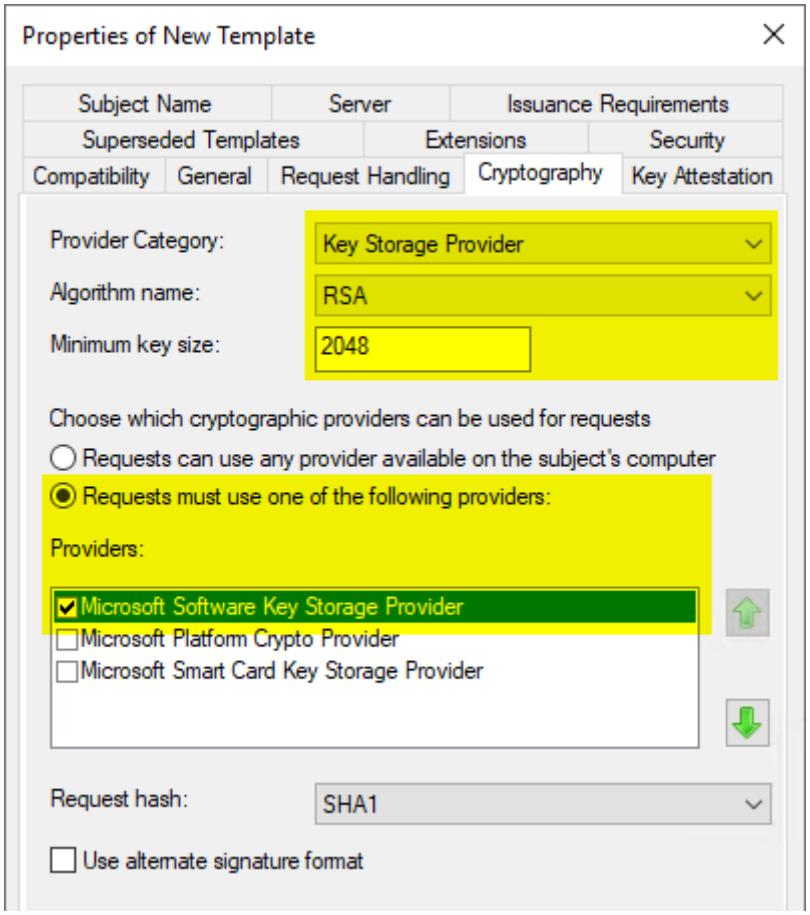
6. In the **Compatibility** tab,
    a. Set **Certification Authority** as **Windows Server 2008 R2**
    b. Set **Certification Recipient** as **Windows 7 / Server 2008 R2**



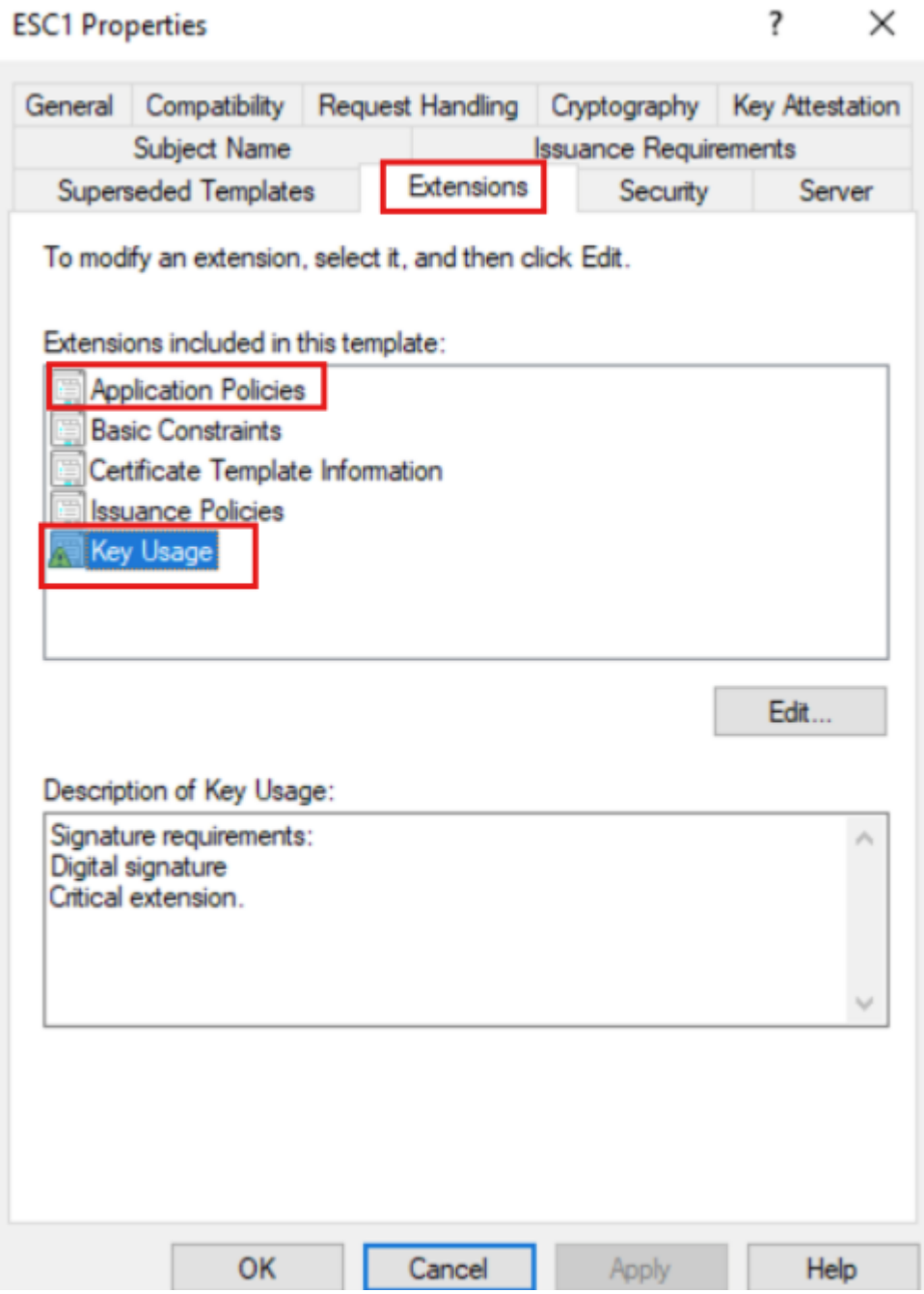
7. In the **Subject Name** tab,
    a. Select **Supply in the request**



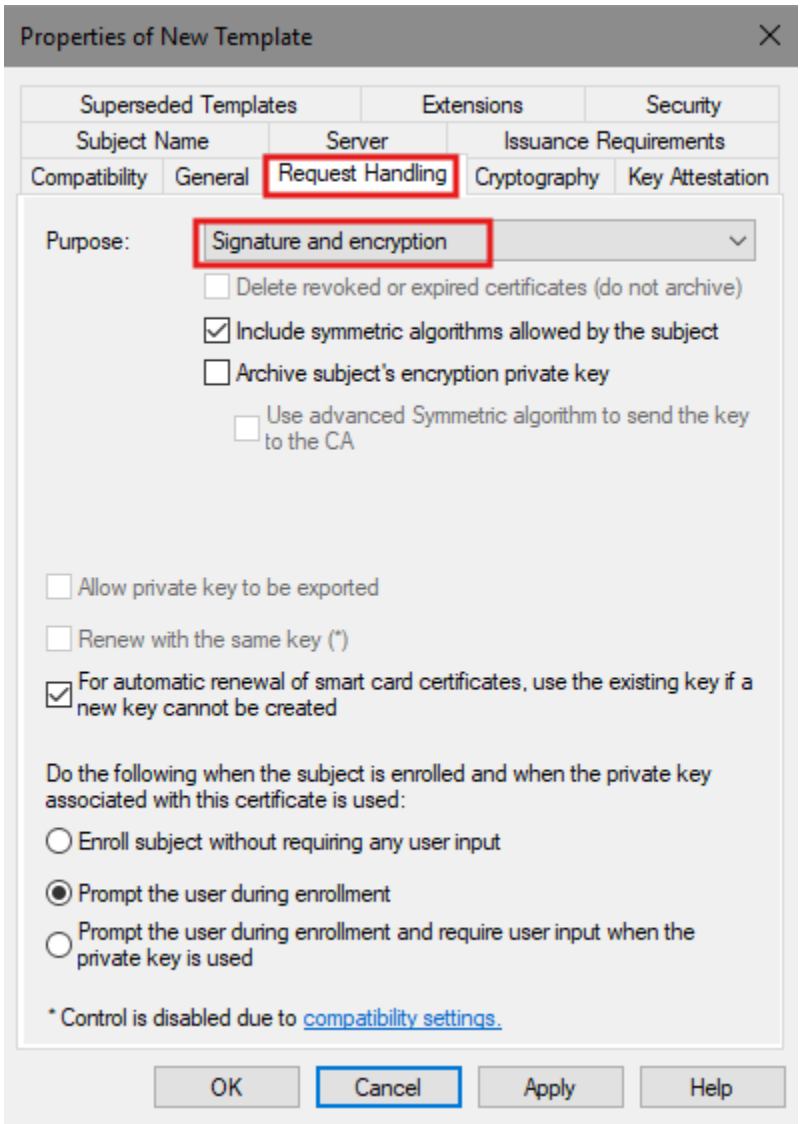
8. In the **Cryptography** tab,
    a. Set **Provider Category** as **Legacy Cryptographic Service Provider**
    b. Set **Algorithm Name** as **RSA**
    c. Set **Minimum Key Size** as **2048**
    d. Check **Requests must use one of the following providers:**
    e. For **Providers**, check **Microsoft RSA Schannel Cryptographic Provider**
    f. For **Request hash**, set to **SHA1**



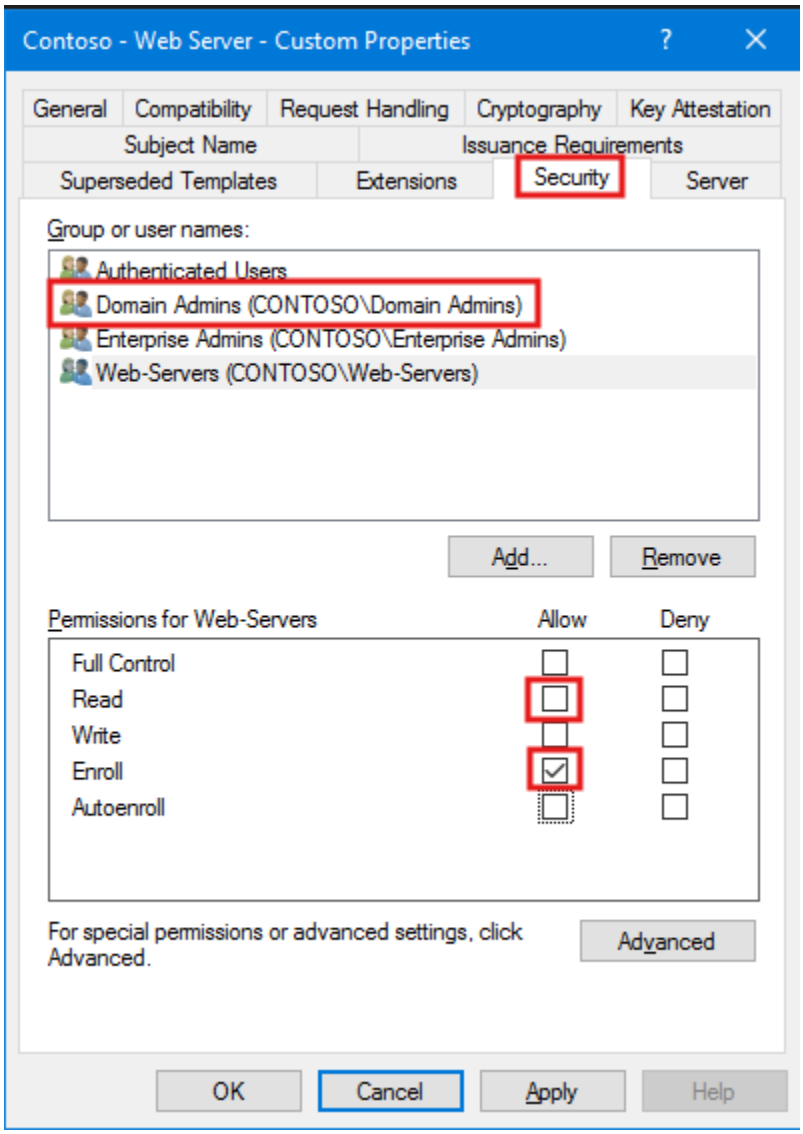
9. In the **Extension** tab,
    a. For **Application Policies**, ensure **Server Authentication** is listed
    b. For **Key Usage**, ensure **Digital Signature** and **Key Encipherment** are listed



10. In the **Request Handling** tab,
    a. Set **Purpose** to **Signature and encryption**
    b. Uncheck **Allow private key to be exported** (for security)



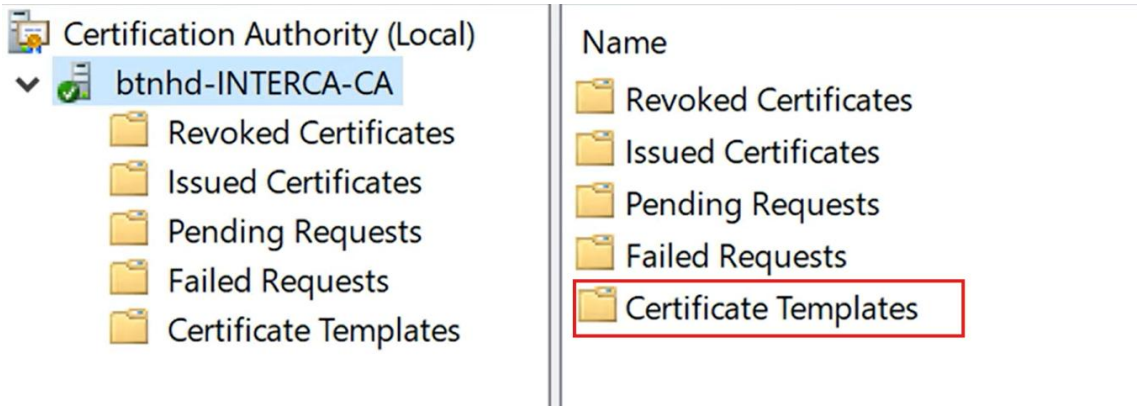
11. In the **Security** tab,
    a. For any group, such as Domain Admins, that will be allowed to enroll/issue certs on InterCA, ensure the permissions, **Read** and **Enroll** are checked to **Allow**



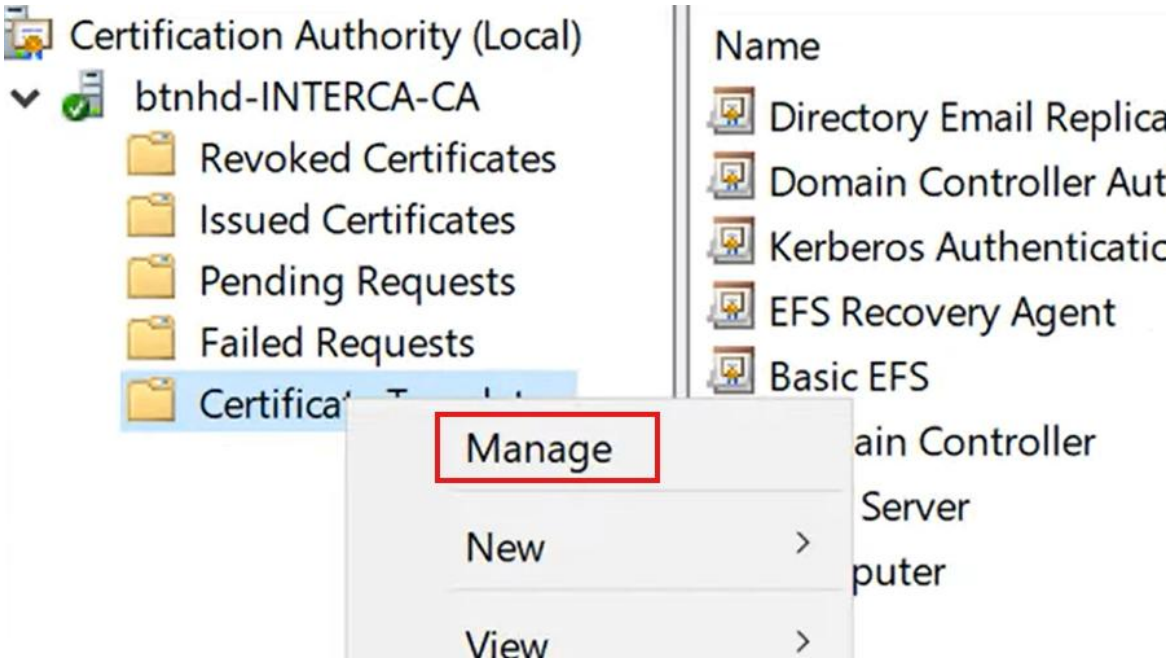
[Low-Security] Creation of SHA1-1024bit Non-Windows Certificate Template

In the Intermediate CA server

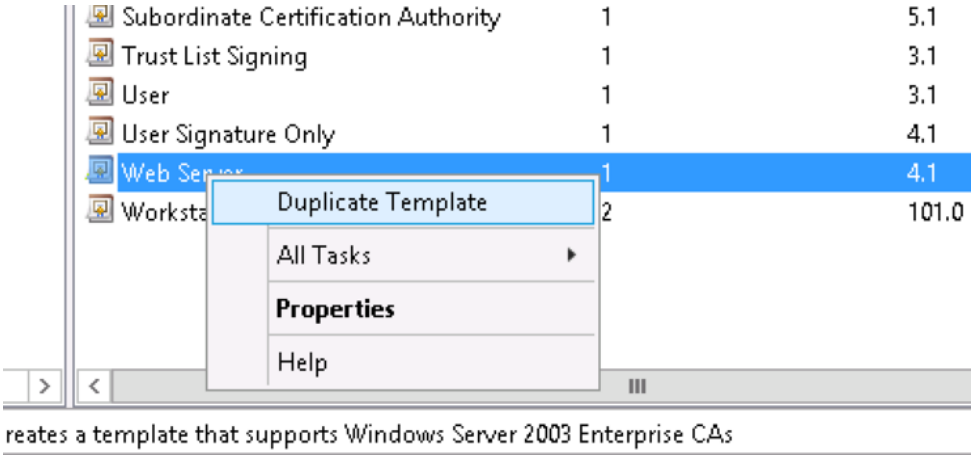
1. Open **Certification Authority (certsrv.msc)**
2. Under our server, go to **Certificate Templates**



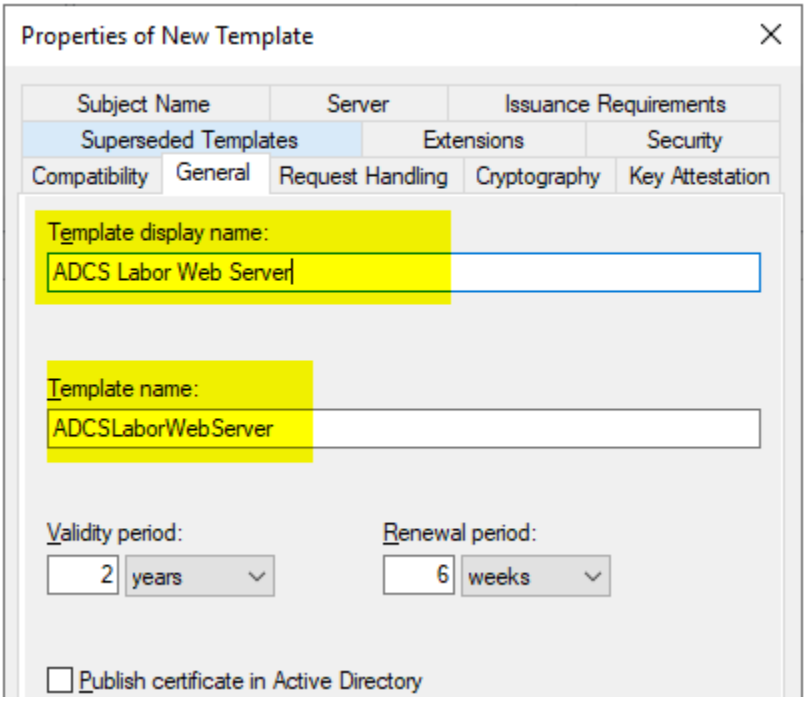
3. Right click **Certificate Templates** and click **Manage**



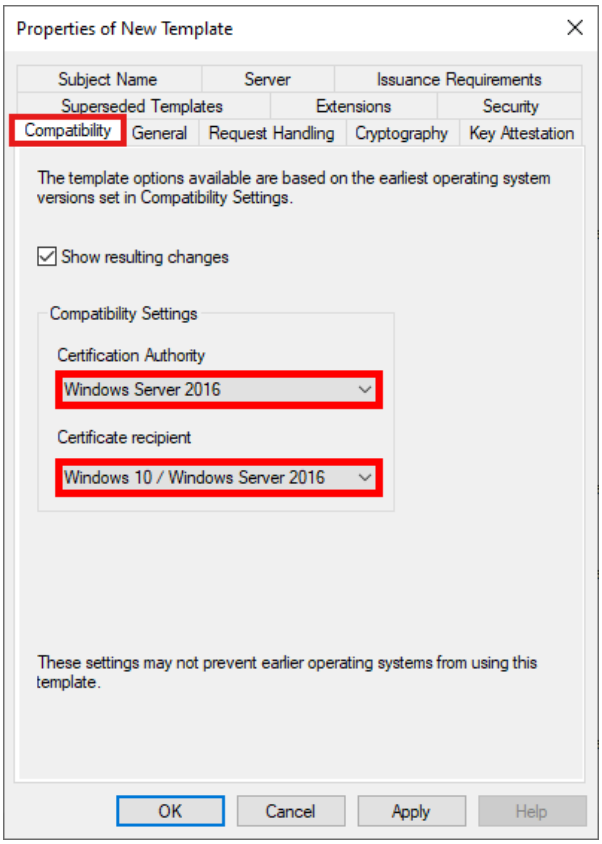
4. Right-click **Web Server** and click **Duplicate Template**



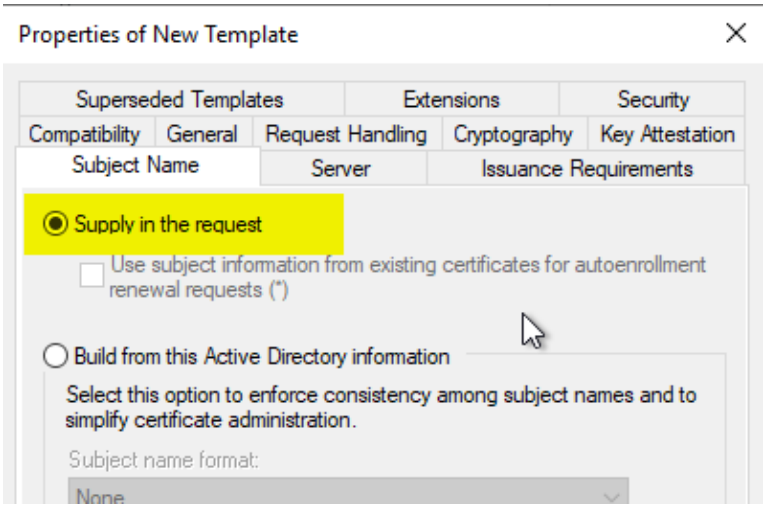
5. In the **General** tab,
   a. Name the template as necessary. I suggest **SHA1-1024 NonWindows Authentication**
   b. Set the **Validity Period** as necessary (preferably **2-3 years**) Some web browsers might alert if validity is longer than 398 days, if so change the Validity Period to 1 year



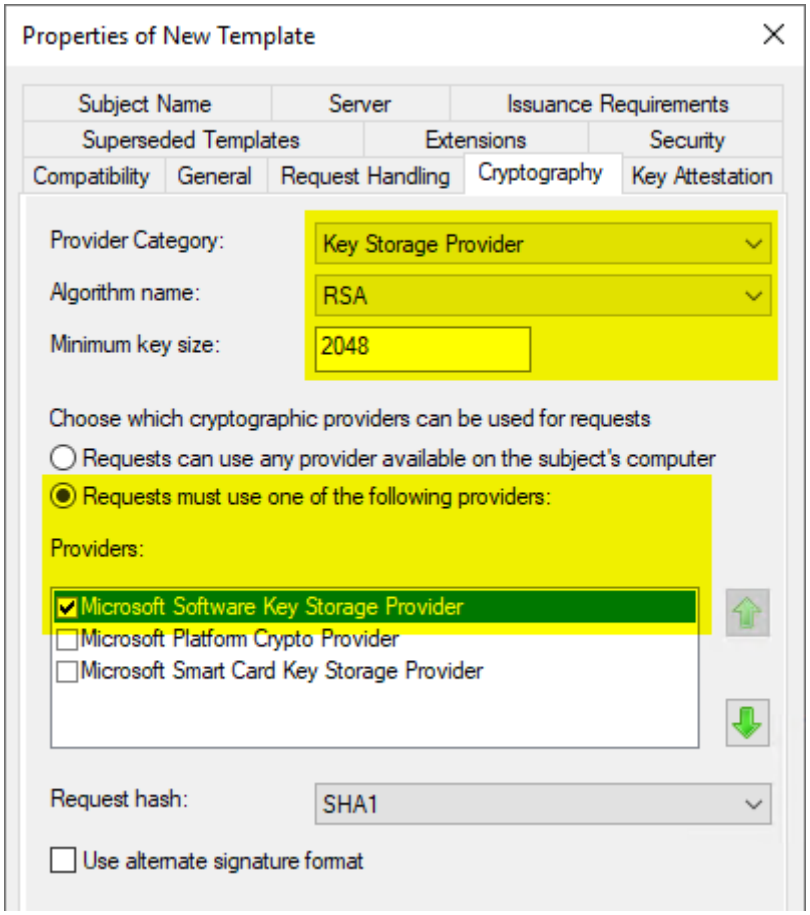
6. In the **Compatibility** tab,
    a. Set **Certification Authority** as **Windows Server 2003**
    b. Set **Certification Recipient** as **Windows XP / Server 2003**



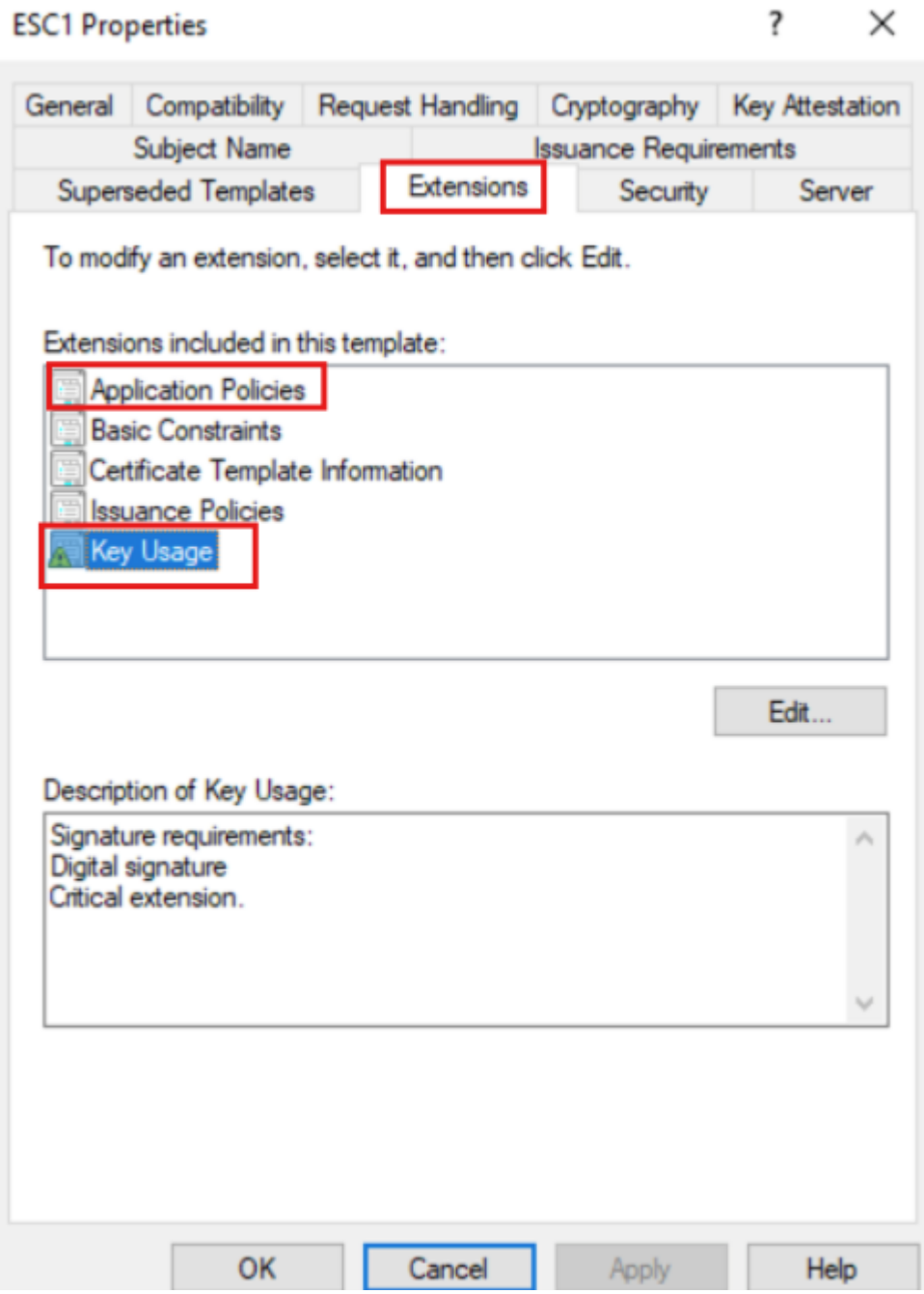
7. In the **Subject Name** tab,
    a. Select **Supply in the request**



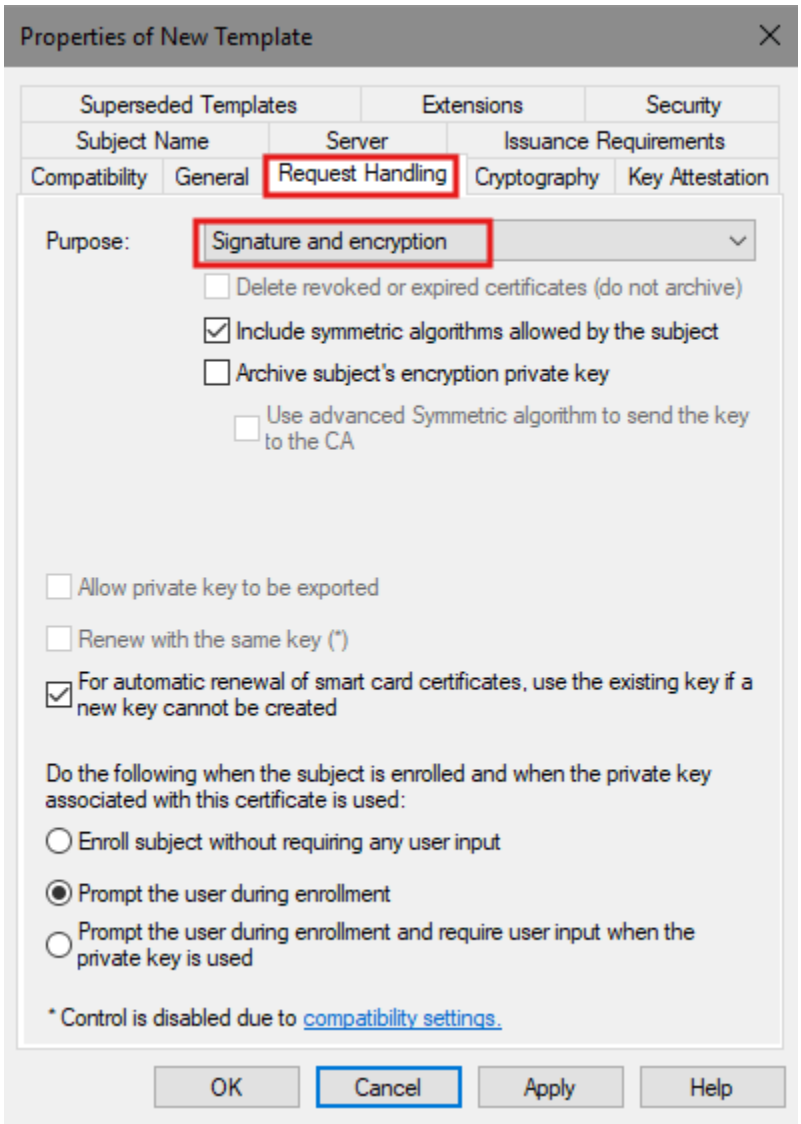
8. In the **Cryptography** tab,
    a. Set **Provider Category** as **Legacy Cryptographic Service Provider**
    b. Set **Algorithm Name** as **RSA**
    c. Set **Minimum Key Size** as **1024**
    d. Check **Requests must use one of the following providers:**
    e. For **Providers**, check **Microsoft Strong Cryptographic Provider**
    f. For **Request hash**, set to **SHA1**



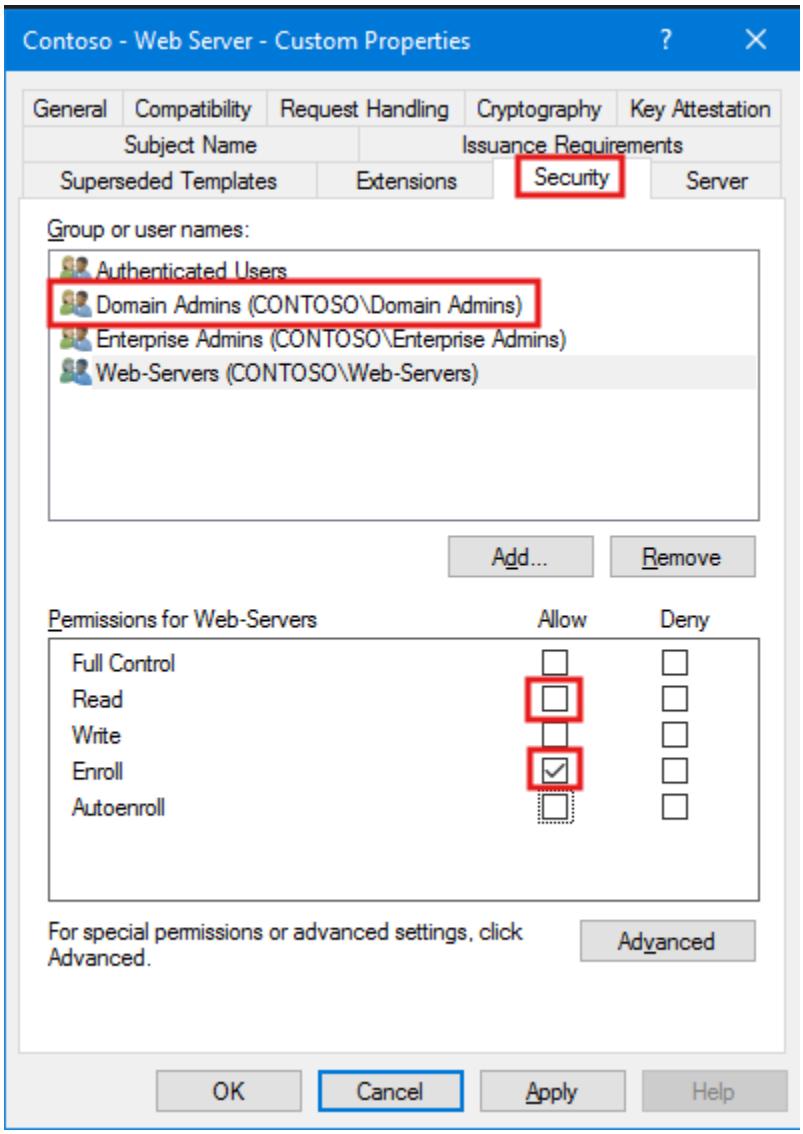
9. In the **Extension** tab,
    c. For **Application Policies**, ensure **Server Authentication** is listed
    d. For **Key Usage**, ensure **Digital Signature** and **Key Encipherment** are listed



10. In the **Request Handling** tab,
    a. Set **Purpose** to **Signature and encryption**
    b. Uncheck **Allow private key to be exported** (for security)



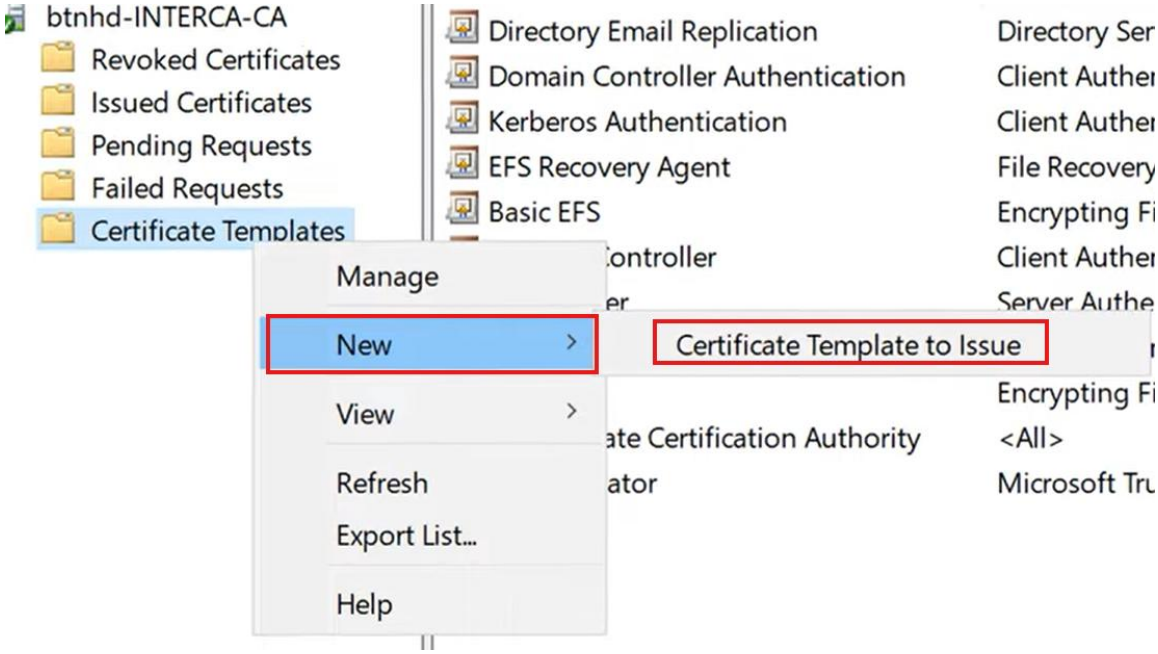
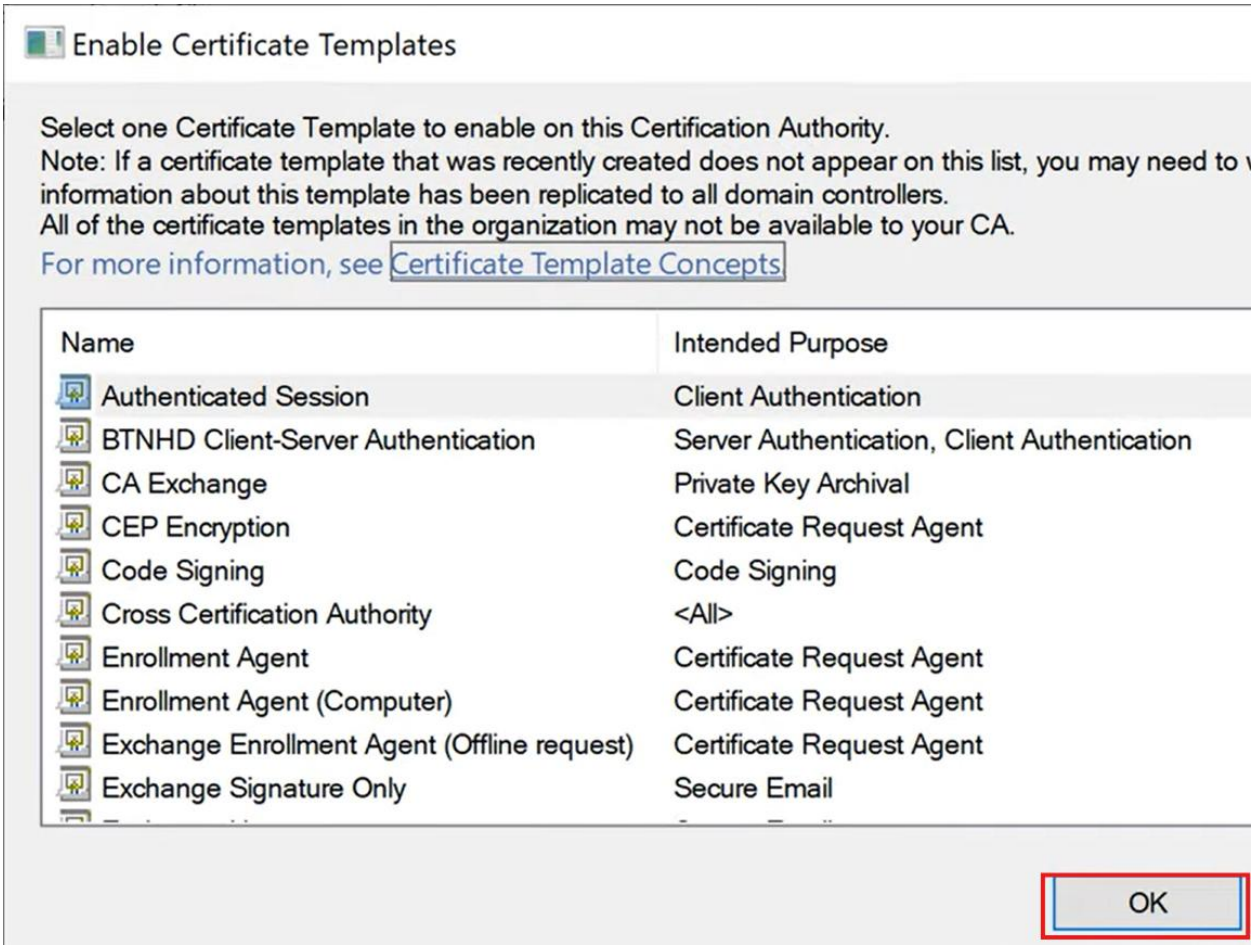
11. In the **Security** tab,
    a. For any group, such as Domain Admins, that will be allowed to enroll/issue certs on InterCA, ensure the permissions, **Read** and **Enroll** are checked to **Allow**



Issuing the newly created Certificate Template

This step needs to be done for each of the three templates created

1. In **Certificate Authority (certsrv.msc)**, under our server, right-click **Certificate Templates**, and click **New > Certificate Template to Issue**



2. In the **Enable Certificate Templates**, select the created certificate template and click **Ok**